



JAIDEV EDUCATION SOCIETY'S  
**J D COLLEGE OF ENGINEERING AND MANAGEMENT**  
KATOL ROAD, NAGPUR

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(An Autonomous Institute, with NAAC "A" Grade)  
Affiliated to DBATU, RTMNU & MSBTE Mumbai  
**Department of Computer Applications**  
*"Powered by Progress, Inspired by Vision"*  
Session 2024-25



**VISION**

"To be the source of bringing out globally competent pioneering computing professionals, researchers, innovators and entrepreneurs and thereby succeed and contribute value to the knowledge-based economy and society."

**MISSION**

1. To offer high-grade, value-based post-graduate programme in the field of Computer Applications.
2. To develop the youth into professionals who can work in team, possess high analytical abilities, and help in solving complex problems of various domains through principles of computer science and applications.

**MCA SEM-III Syllabus**

| Sr No | Category | Course Code | Subject                            | L  | T | P  | CA  | ESE | Total | Credit |
|-------|----------|-------------|------------------------------------|----|---|----|-----|-----|-------|--------|
| 1     | PCC      | MC3T001     | Cloud Computing                    | 4  | 0 | 0  | 40  | 60  | 100   | 4      |
| 2     | PCC      | MC3T002     | BlockChain Technology              | 3  | 0 | 0  | 40  | 60  | 100   | 3      |
| 3     | PCC      | MC3T003     | Machine Learning                   | 3  | 0 | 0  | 40  | 60  | 100   | 3      |
| 4     | PCC      | MC3T004     | R Programming for Data Science     | 3  | 0 | 0  | 40  | 60  | 100   | 3      |
| 5     | PEC      | MC3E00*     | Professional Elective-II           | 3  | 0 | 0  | 40  | 60  | 100   | 3      |
| 6     | PCC      | MC3L006     | Machine Learning LAB               | 0  | 0 | 4  | 60  | 40  | 100   | 2      |
| 7     | PCC      | MC3P007     | Mini Project Seminar               | 0  | 0 | 4  | 60  | 40  | 100   | 2      |
| 8     | PCC      | MC3L008     | R Programming for Data Science LAB | 0  | 0 | 4  | 60  | 40  | 100   | 2      |
| Total |          |             |                                    | 16 | 0 | 12 | 380 | 420 | 800   | 22     |

| Semester | Course Code | Name of the course | L | T | P | Credits |
|----------|-------------|--------------------|---|---|---|---------|
| III      | MC3T001     | CLOUD COMPUTING    | 4 | 0 | 0 | 4       |

| Prerequisites for the course |                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                           | To provide students with the fundamentals and essentials of Cloud Computing.                                                                                                     |
| 2.                           | To provide students a sound foundation of the Cloud Computing so that they are able to start using and adopting Cloud Computing services and tools in their real life scenarios. |

| Prior Reading Material/useful links |                                                                                                                                                                         |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                                  | <a href="https://cloud.google.com/learn/what-is-cloud-computing">https://cloud.google.com/learn/what-is-cloud-computing</a>                                             |
| 2.                                  | <a href="https://www.javatpoint.com/cloud-computing">https://www.javatpoint.com/cloud-computing</a>                                                                     |
| 3                                   | <a href="https://www.salesforce.com/products/platform/best-practices/cloud-computing/">https://www.salesforce.com/products/platform/best-practices/cloud-computing/</a> |

#### Course Outcomes:

| Sr. No | Course Outcome | CO statement                                                                                                                                                                                                              |
|--------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | CO1            | Understand the core concepts of the cloud computing and its benefits along with its various models and services in cloud computing                                                                                        |
| 2      | CO2            | Explain the core concepts of the cloud computing paradigm: how and why this paradigm shift came about, the characteristics, advantages and challenges brought about by the various models and services in cloud computing |
| 3      | CO3            | Apply the fundamental concepts in datacenters to understand the tradeoffs in power, efficiency and cost.                                                                                                                  |
| 4      | CO4            | Identify resource management fundamentals, i.e. resource abstraction, sharing and sandboxing and outline their role in managing infrastructure in cloud computing.                                                        |
| 5      | CO5            | Analyze various cloud programming models and apply them to solve problems on the cloud.                                                                                                                                   |

**Syllabus:**

| <b>Course Contents</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Hours</b>   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Unit I                 | Cloud Computing Fundamental: History of cloud computing, Cloud Computing definition, private, public and hybrid cloud. Applications and challenges of cloud computing. Types of Cloud Services: IaaS, PaaS, SaaS., Public Cloud Vs Private Clouds.                                                                                                                                                                                                                                                                                                                  | <b>[8 Hrs]</b> |
| Unit II                | Cloud Architecture: Introduction to Architecture, Benefits and challenges, Application availability, performance, security and disaster recovery; future of Cloud Applications. Desktop and Device Management: Introduction-Objectives, Desktop Virtualization- Across Industries Client Desktops, Desktop placement in the cloud Merits Desktop as a Service (DaaS), Desktop Management Watching the four areas Asset Management.                                                                                                                                  | <b>[7 Hrs]</b> |
| Unit III               | Virtualization: Introduction to Virtualization, Network virtualization techniques, Virtual Machine (VM), VM Components and process of converting physical to VMs, Block virtualization and file level storage virtualization, Virtual LAN (VLAN) and Virtual SAN (VSAN).                                                                                                                                                                                                                                                                                            | <b>[7 Hrs]</b> |
| Unit IV                | Cloud Application Development: Service creation environments, Development environments, Amazon, Azure, Google App. Cloud Applications: Technologies and the processes required when deploying web services; Deploying a web service from inside and outside a cloud architecture, advantages and disadvantages. Accessing the Cloud Introduction-Objectives, Platforms Web Application Framework- Web Hosting Services Proprietary Methods, Web Applications APIs in Cloud Computing, Browsers for Cloud Computing Internet Explorer Mozilla Firefox Safari Chrome. | <b>[7 Hrs]</b> |
| Unit V                 | Cloud Security: Cloud Security Overview, Cloud Security Challenges and Secure Cloud Software Requirements. Risks: Risk Management, Privacy and compliance risk. Software-as-a-Service Security, Security Governance, Security Monitoring, Security Architecture Design. Data Security, Application Security, Virtual Machine Security, Identity Management and Access Control, Autonomic Security.                                                                                                                                                                  | <b>[7 Hrs]</b> |

**Text Books**

|    |                                                                                                        |
|----|--------------------------------------------------------------------------------------------------------|
| 1. | Toby Velte, Anthony Velte, Robert Elsenpeter, Cloud Computing, A Practical Approach [ISBN: 0071626948] |
| 2. | Rajkumar Buyya, Christian Vecchiola, S.TamaraiSelvi, Mastering Cloud Computing, TMGH, 2013.            |
| 3. | Barrie Sosinsky: "Cloud Computing Bible", Wiley-India, 2010                                            |

**Reference Books**

|    |                                                                                                                        |
|----|------------------------------------------------------------------------------------------------------------------------|
| 1. | GautamShroff, Enterprise Cloud Computing Technology Architecture Applications                                          |
| 2. | Ronald L. rutz, Russell Dean Vines, Cloud Security A comprehensive Guide to Secure Cloud Computing, Wiley India, 2010. |
| 3. | John W.Rittinghouse and Ames F.Ransome, Cloud Computing: Implementation, Management, and Security, CRC Press, 2010.    |
| 4. | Kumar Saurabh, Cloud Computing insights into New-Era Infrastructure, Wiley India,2011.                                 |

| <b>Useful Links</b> |                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1.                  | <a href="https://onlinecourses.nptel.ac.in/noc22_cs91/preview">https://onlinecourses.nptel.ac.in/noc22_cs91/preview</a> |
| 2.                  | <a href="https://github.com/topics/mini-project">https://github.com/topics/mini-project</a>                             |

| Semester | Course Code | Name of the course     | L | T | P | Credits |
|----------|-------------|------------------------|---|---|---|---------|
| III      | MC3T002     | BLOCK CHAIN TECHNOLOGY | 3 | 0 | 0 | 3       |

| Prerequisites for the course |                                                       |
|------------------------------|-------------------------------------------------------|
| 1.                           | Vulnerabilities in the Information Technology systems |
| 2.                           | Anticipating and detecting threats                    |
| 3.                           | Routing, switching and firewall                       |
| 4.                           | Being aware of the network architecture and protocol  |

| Prior Reading Material/useful links |                                                                                                                                                                                               |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                                  | <a href="http://nptel.ac.in/courses/106105031/lecture">http://nptel.ac.in/courses/106105031/lecture</a> by Dr.Debdeep Mukhopadhyay IIT Kharagpur                                              |
| 2.                                  | <a href="https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-033-computer-">https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-033-computer-</a> |

#### Course Outcomes:

| Sr. No | Course Outcome number | CO statement                                                                                                                                                                                         |
|--------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | CO1                   | Understand emerging abstract models for Blockchain Technology.                                                                                                                                       |
| 2      | CO2                   | Identify major research challenges and technical gaps existing between theory and practice in crypto currency domain.                                                                                |
| 3      | CO3                   | Provides conceptual understanding of the function of Blockchain as a method of securing distributed ledgers, how consensus on their contents is achieved, and the new applications that they enable. |
| 4      | CO4                   | Apply hyperledger Fabric and Ethereum platform to implement the Block chain Application                                                                                                              |
| 5      | CO5                   | Apply appropriate Intrusion detection and prevention techniques and to examine various security algorithms to Interpret security incidents                                                           |

**Syllabus:**

| <b>Course Contents</b> |                                                                                                                                                                                                                                                                                                                                                                | <b>Hours</b>   |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Unit I                 | Blockchain- Public Ledgers, Blockchain as Public Ledgers -Bitcoin, Blockchain 2.0, Smart Contracts, Block in a Blockchain, Transactions-Distributed Consensus, The Chain and the Longest Chain - Cryptocurrency to Blockchain 2.0 - Permissioned Model of Blockchain, Cryptographic -Hash Function, Properties of a hash function-Hash pointer and Merkle tree | <b>[7 Hrs]</b> |
| Unit II                | A basic crypto currency, Creation of coins, Payments and double spending, FORTH – the precursor for Bitcoin scripting, Bitcoin Scripts , Bitcoin P2P Network, Transaction in Bitcoin Network, Block Mining, Block propagation and block relay, Consensus introduction, Distributed consensus in open environments-Consensus in a Bitcoin network               | <b>[7 Hrs]</b> |
| Unit III               | Bitcoin Consensus, Proof of Work (PoW)- Hashcash PoW , Bitcoin PoW, Attacks on PoW ,monopoly problem- Proof of Stake- Proof of Burn - Proof of Elapsed Time - Bitcoin Miner, Mining Difficulty, Mining Pool-Permissioned model and use cases, Design issues for Permissioned Blockchains                                                                       | <b>[7 Hrs]</b> |
| Unit IV                | RAFT Consensus-Byzantine general problem, Byzantine fault tolerant system-Agreement Protocol, Lamport-Shostak-Pease BFT Algorithm-BFT over Asynchronous systems, Practical Byzantine Fault Tolerance                                                                                                                                                           | <b>[8 Hrs]</b> |
| Unit V                 | Architecture of Hyperledger fabric v1.1-Introduction to hyperledger fabric v1.1, chain code- Ethereum: Ethereum network, EVM, Transaction fee, Mist Browser, Ether, Gas, Solidity, Smart contracts, TruffleDesign and issue Crypto currency, Mining, DApps, DAO                                                                                                | <b>[7 Hrs]</b> |

| <b>Text Books</b> |                                                                                                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                | Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder. Bitcoin and cryptocurrency technologies: a comprehensive introduction. Princeton University Press, 2016 |
| 2.                | Draft version of “S. Shukla, M. Dhawan, S. Sharma, S. Venkatesan, ‘Blockchain Technology: Cryptocurrency and Applications’, Oxford University Press, 2019.                                    |

| <b>Reference Books</b> |                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                     | Mastering Blockchain: Deeper insights into decentralization, cryptography, Bitcoin, and popularBlockchain frameworks by Bashir, Imran,2017. |
| 2.                     | Joseph Bonneauetalsok: Research perspectives and challenges for Bitcoin andcryptocurrency, IEEE Symposium on security and P                 |

| <b>Useful Links</b> |                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                  | <a href="http://nptel.ac.in/courses/106105031/lecture">http://nptel.ac.in/courses/106105031/lecture</a> by Dr.Debdeep Mukhopadhyay IIT Kharagpur                          |
| 2.                  | <a href="https://ocw.mit.edu/courses/-engineering-and-computer-science/6-033-computer-">https://ocw.mit.edu/courses/-engineering-and-computer-science/6-033-computer-</a> |



| Semester                            | Course Code                                                                                                                                                                                                                     | Name of the course | L | T | P | Credits |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---|---|---|---------|
| III                                 | MC3T003                                                                                                                                                                                                                         | MACHINE LEARNING   | 3 | 0 | 0 | 3       |
|                                     | Prerequisites for the course                                                                                                                                                                                                    |                    |   |   |   |         |
| 1                                   | Capability to implement basic algorithms using basic machine learning applying ML to problems encountered in various domains. In addition, obtain exposure to high-level MLlibraries or frameworks such as TensorFlow, PyTorch. |                    |   |   |   |         |
| 2                                   | Make aware of the role of data in the future of computing, and also in solving real-world problems using machine learning algorithms.                                                                                           |                    |   |   |   |         |
| 3.                                  | Help connect real-world problems to appropriate ML algorithm(s) for solving them. Enable formulating real world problems as machine learning tasks.                                                                             |                    |   |   |   |         |
| Prior Reading Material/useful links |                                                                                                                                                                                                                                 |                    |   |   |   |         |
| 1                                   | https://www.shahucollegelatur.org.in/Department/Studymaterial/sci/it/BCA/FY/digielec.pdf                                                                                                                                        |                    |   |   |   |         |
| 2                                   | https://byjusexamprep.com/best-books-for-digital-circuits-i-0059fcd0-7e8f-11e7-b373-b30de31b2e0e                                                                                                                                |                    |   |   |   |         |
| 3                                   | https://www.sanfoundry.com/best-reference-books-basics-digital-electronics/                                                                                                                                                     |                    |   |   |   |         |

**Course Outcomes:**

| Sr. No | Course Outcome number | CO statement                                                                     |
|--------|-----------------------|----------------------------------------------------------------------------------|
| 1      | CO1                   | Demonstrate knowledge of the machine learning literature.                        |
| 2      | CO2                   | Describe how and why machine learning methods work.                              |
| 3      | CO3                   | Demonstrate results of parameter selection.                                      |
| 4      | CO4                   | Explain relative strengths and weaknesses of different machine learning methods. |
| 5      | CO5                   | Apply appropriate machine learning methods to a selected problem.                |



**Syllabus:**

| Course Contents |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Hours  |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Unit I          | <b>Introduction to Machine Learning</b> Machine Learning, Types of Machine Learning, Issues in Machine Learning, Application of Machine Learning, Steps in developing a Machine Learning Application.                                                                                                                                                                                                                                                                                                                             | [7Hrs] |
| UNIT II         | <b>Introduction to Neural Network</b> Introduction – Fundamental concept – Evolution of Neural Networks Biological Neuron, Artificial Neural Networks, NN architecture, Activation functions, McCulloch-Pitts Model.                                                                                                                                                                                                                                                                                                              | [7Hrs] |
| Unit III        | <b>Introduction to Optimization Techniques:</b> Derivative based optimization- Steepest Descent, Newton method. Derivative free optimization- Random Search, Down Hill Simplex                                                                                                                                                                                                                                                                                                                                                    | [8Hrs] |
| Unit IV         | <b>Learning with Regression and trees:</b> Learning with Regression : Linear Regression, Logistic Regression. Learning with Trees: Decision Trees, Constructing Decision Trees using Gini Index, Classification and Regression Trees (CART).                                                                                                                                                                                                                                                                                      | [7Hrs] |
| Unit V          | <b>Learning with Classification and clustering:</b><br><b>Classification:</b> Rule based classification, classification by Bayesian Belief networks, Hidden Markov Models. <b>Support Vector Machine:</b> Maximum Margin Linear Separators<br><b>Clustering:</b> Expectation Maximization Algorithm, Supervised learning after clustering, Radial Basis functions. <b>Dimensionality Reduction:</b> Dimensionality Reduction Techniques, Principal Component Analysis, Independent Component Analysis, Single value decomposition | [7Hrs] |

| Text Books |                                                                                                                         |
|------------|-------------------------------------------------------------------------------------------------------------------------|
| 1.         | Mitchell, Tom. M., “Machine Learning”, McGraw-Hill Education,                                                           |
| 2.         | Segaran, Toby. “Programming Collective Intelligence- Building Smart Web 2.0 Applications”, O’Reilly Media, August 2007. |

| Reference Books |                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------|
| 1.              | Miroslav, Kubat. “An Introduction to Machine Learning”, Springer Publishing.                      |
| 2.              | Bishop, C. M., “Pattern Recognition and Machine Learning”, Springer Publishing.                   |
| 3.              | Conway, Drew and White, John Myles, “Machine Learning for Hackers”, O’Reilly Media, February 2012 |

| Useful Links |                                                                                                                                                                                                                       |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://www.youtube.com/results?search_query=digital+electronics+">https://www.youtube.com/results?search_query=digital+electronics+</a>                                                                     |
| 2            | <a href="https://www.youtube.com/watch?v=M0mx8S05v60&amp;list=PLBlnK6fEyqRjMH3mWf6kwqiTbT798eAOm">https://www.youtube.com/watch?v=M0mx8S05v60&amp;list=PLBlnK6fEyqRjMH3mWf6kwqiTbT798eAOm</a>                         |
| 3            | <a href="https://www.youtube.com/watch?v=izBaDRyqnBk&amp;list=PLBlnK6fEyqRjMH3mWf6kwqiTbT798eAOm&amp;index=4">https://www.youtube.com/watch?v=izBaDRyqnBk&amp;list=PLBlnK6fEyqRjMH3mWf6kwqiTbT798eAOm&amp;index=4</a> |

| Semester | Course Code | Name of the course             | L | T | P | Credits |
|----------|-------------|--------------------------------|---|---|---|---------|
| III      | MC3T003     | R Programming for Data Science | 3 | 0 | 0 | 3       |

#### Prerequisites for the course

|   |                                               |
|---|-----------------------------------------------|
| 1 | Knowledge of C Programming and Data Structure |
|---|-----------------------------------------------|

#### Prior Reading Material/useful links

|   |                                                                                                                                                                                                                         |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Tutorials Point (I) simply easy learning, Online Tutorial Library (2018), R Programming, Retrieved from <a href="https://www.tutorialspoint.com/r/r_tutorial.pdf">https://www.tutorialspoint.com/r/r_tutorial.pdf</a> . |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Course Objectives

| Sr. No | Statement                                            |
|--------|------------------------------------------------------|
| 1      | Understanding the Basics of R and R Studio           |
| 2      | Mastering Data Types, Variables, and Operators in R  |
| 3      | Implementing Control Structures and Functions in R   |
| 4      | Manipulating Data with R Lists, Matrices, and Arrays |
| 5      | Working with Data Frames and Data Manipulation in R  |

#### Course Outcomes

| Sr. No | Code | CO statement                                            |
|--------|------|---------------------------------------------------------|
| 1      | CO1  | Understand the Basics of R Programming and Data Science |
| 2      | CO2  | Apply Data Types and Structures in R                    |
| 3      | CO3  | Develop Functions and Manipulate Text Data in R         |
| 4      | CO4  | Analyze and Compute Matrix and Array Operations         |
| 5      | CO5  | Construct and Manage Data Frames in R                   |

| Unit No | Course Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Duration |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| I       | <p><b>Introduction to Data Science:</b> Definition and scope of Data Science, Importance of Data Science in various fields (finance, healthcare, social media, etc.)</p> <p><b>Data Science lifecycle:</b> Data collection, Data cleaning, Data exploration, Data modelling, and Interpretation.</p> <p><b>Roles in Data Science:</b> Data Scientist, Data Engineer, Analyst.</p> <p><b>Introduction to R:</b> What is R? – Why R? – Advantages of R over Other Programming Languages - R Studio: R command Prompt, R script file, comments –Installing a R Package, Special Values functions : NA, Inf and -inf.</p> | 6 hours  |
| II      | <p><b>R Data Types:</b> Vectors, Lists, Matrices, Arrays, Factors, Data Frame – R - Variables: Variable assignment, Data types of Variable, Finding Variable ls(), Deleting Variables – R Operators: Arithmetic Operators, Relational Operators, Logical Operator, Assignment Operators, Miscellaneous Operators –</p>                                                                                                                                                                                                                                                                                                | 7 hours  |

|            |                                                                                                                                                                                                                                                                                                                                                |         |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|            | <b>R Decision Making:</b> if statement, if – else statement, if– else if statement, switch statement – R Loops: repeat loop, while loop, for loop – Loop control statement: break statement, next statement.                                                                                                                                   |         |
| <b>III</b> | <b>R-Function :</b> function definition, Built in functions: mean(), paste(), sum(), min(), max(), seq(), user-defined function, calling a function, calling a function without an argument, calling a function with argument values - RStrings – <b>Manipulating Text in Data:</b> substr(),strsplit(), paste(), grep(), toupper(), tolower() | 7 hours |
| <b>IV</b>  | <b>List:</b> Size of List, Merging Lists, Converting List to Vector - <b>R Matrices</b> – Accessing Elements of a Matrix, Matrix Computations: Addition, subtraction, Multiplication and Division<br><b>R Arrays:</b> Naming Columns and Rows, Accessing Array Elements, Manipulating Array Elements,                                          | 8 hours |
| <b>V</b>   | <b>Data Frames</b> –Create Data Frame, Data Frame Access, Understanding Data in Data Extract Data from Data Frame, Expand Data Frame: Add Column, Add Row - Joining columns and rows in a Data frame rbind() and cbind() – Merging Data frames merge().                                                                                        | 8 hours |

| Text Books |                             |                                         |                               |                           |
|------------|-----------------------------|-----------------------------------------|-------------------------------|---------------------------|
| Code       | Title of the Book           | Author Name /Designation / Organization | Publisher                     | Edition/ Publication Year |
| T1         | R Programming for Beginners | Sandip Rakshit                          | McGraw Hill Education (India) | 2017                      |
| T2         | Data Analytics using R      | Seema Acharya                           | McGrawHill Education (India)  | 2018                      |

| Reference Books |                             |                                       |                          |                           |
|-----------------|-----------------------------|---------------------------------------|--------------------------|---------------------------|
| Code            | Title of the Book           | Author Name/Designation/ Organization | Publisher                | Edition/ Publication Year |
| T1              | R for Dummies A Wiley Brand | Andrie de Vries, Joris Meys           | John Wileyand Sons, Inc, | 2nd Edition/2015          |

| Useful Links |                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------|
| 1            | <a href="https://coursera.org/learn/r-programming">https://coursera.org/learn/r-programming</a> |
| 2            | <a href="https://www.r-project.org/about.html">https://www.r-project.org/about.html</a>         |
| 3            | <a href="https://www.programiz.com/r">https://www.programiz.com/r</a>                           |

| Semester | Course Code | Name of the course                              | L | T | P | Credits |
|----------|-------------|-------------------------------------------------|---|---|---|---------|
| III      | MC3E001     | PROFESSIONAL<br>ELECTIVE-II-SOFTWARE<br>TESTING | 3 | 0 | 0 | 3       |

#### Prerequisites for the course

|    |                                                                                                                                  |
|----|----------------------------------------------------------------------------------------------------------------------------------|
| 1. | The main goal of software testing is to find bugs as early as possible and fix bugs and make sure that the software is bug-free. |
|----|----------------------------------------------------------------------------------------------------------------------------------|

#### Prior Reading Material/useful links

|    |                                                                                                                             |
|----|-----------------------------------------------------------------------------------------------------------------------------|
| 1. | <a href="https://www.geeksforgeeks.org/software-testing-basics/">https://www.geeksforgeeks.org/software-testing-basics/</a> |
| 2. | <a href="https://www.javatpoint.com/software-testing-tutorial">https://www.javatpoint.com/software-testing-tutorial</a>     |

#### Course Outcomes:

| Sr. No | Course Outcome number | CO statement                                                                                                                                  |
|--------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | CO1                   | Understand different software testing techniques and strategies and be able to apply specific(automated) unit testing method to the projects. |
| 2      | CO2                   | Distinguish characteristics of structural testing methods.                                                                                    |
| 3      | CO3                   | Demonstrate the integration testing which aims to uncover interaction and compatibility problems as early as possible.                        |
| 4      | CO4                   | Discuss about the functional and system testing methods.                                                                                      |
| 5      | CO5                   | Demonstrate various issues for object oriented testing.                                                                                       |

**Syllabus:**

| <b>Course Contents</b> |                                                                                                                                                                                                                                             | <b>Hours</b>   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Unit I-                | Mathematical Context: Perspective on Testing, Examples Functional Testing: Boundary Value Testing, Equivalence Class Testing, Decision TableBased Testing, Retrospective on Functional Testing                                              | <b>[7 Hrs]</b> |
| Unit II-               | Structural Testing: Path Testing- DD-Paths, Test Coverage Metrics, Basis Path Testing, Dataflow Testing- Define/Use Testing, Slice-Based Testing, Retrospective on Structural Testing- Gaps and Redundancies, Metrics for Method Evaluation | <b>[7 Hrs]</b> |
| Unit III               | Structural Testing: Path Testing- DD-Paths, Test Coverage Metrics, Basis Path Testing, Dataflow Testing- Define/Use Testing, Slice-Based Testing, Retrospective on Structural Testing- Gaps and Redundancies, Metrics for Method Evaluation | <b>[7 Hrs]</b> |
| Unit IV                | Threads, Basic Concepts for Requirements Specification, Finding Threads, Structural Strategies for Thread Testing, Functional Strategies for Thread Testing SATM Test Threads, System Testing Guidelines                                    | <b>[8 Hrs]</b> |
| Unit V                 | Object-Oriented Testing: Issues in Object-Oriented Testing, Class Testing, Object-Oriented Integration Testing, GUI Testing, Object-Oriented System Testing                                                                                 | <b>[7 Hrs]</b> |

**Text Books**

|    |                                                                                            |
|----|--------------------------------------------------------------------------------------------|
| 1. | Paul C. Jorgensen, Software Testing: A Craftsman's Approach, 3rd Edition, CRC Press, 2007. |
|----|--------------------------------------------------------------------------------------------|

**Reference Books**

|    |                                                            |
|----|------------------------------------------------------------|
| 1. | Boris Beizer, Software Testing Techniques, Dreamtech, 2009 |
|----|------------------------------------------------------------|

| Semester | Course Code | Name of the course                                 | L | T | P | Credits |
|----------|-------------|----------------------------------------------------|---|---|---|---------|
| III      | MC3E002     | PROFESSIONAL<br>ELECTIVE-II- BIG DATA<br>ANALYTICS | 3 | 0 | 0 | 3       |

#### Prerequisites for the course

|    |                                                                                                                           |
|----|---------------------------------------------------------------------------------------------------------------------------|
| 1. | Knowledge of Procedural an effort that's slower and less efficient with more traditional business intelligence solutions. |
|----|---------------------------------------------------------------------------------------------------------------------------|

#### Prior Reading Material/useful links

|    |                                                                                                                                                                                             |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | <a href="https://www.udemy.com/course/relational-database-management-systemrdbms-complete-pack/">https://www.udemy.com/course/relational-database-management-systemrdbms-complete-pack/</a> |
| 2. | <a href="https://www.coursera.org/learn/database-management">https://www.coursera.org/learn/database-management</a>                                                                         |
| 3  | <a href="https://onlinecourses.nptel.ac.in/noc22_cs91/preview">https://onlinecourses.nptel.ac.in/noc22_cs91/preview</a>                                                                     |
| 4  | <a href="https://www.simplilearn.com/what-is-database-management-article">https://www.simplilearn.com/what-is-database-management-article</a>                                               |

#### Course Outcomes:

| Sr. No | Course Outcome number | CO statement                                                                               |
|--------|-----------------------|--------------------------------------------------------------------------------------------|
| 1      | CO1                   | Outline the basic big data concept                                                         |
| 2      | CO2                   | Categorize and summarize the processing in Big Data and its importance.                    |
| 3      | CO3                   | Simulate various Big data technologies like Hadoop MapReduce, Pig, Hive, Hbase and No- SQL |
| 4      | CO4                   | Determine tools and techniques to analyze Big Data                                         |
| 5      | CO5                   | Resolve problems associated with big data with the features of R programming               |

**Syllabus:**

| <b>Course Contents</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Hours</b> |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Unit I-                | FUNDAMENTALS OF BIG DATA Introduction to big data - convergence of key trends – Applications of big data in web analytics, marketing, fraud, risk, credit risk management, algorithmic trading, advertising, healthcare and medicine – big data technologies – introduction to open source technologies – Hadoop, cloud and big data – mobile business intelligence – Crowd sourcing analytics – inter and trans firewall analytics - Information Management .                                                                                                                                              | [7 Hrs]      |
| Unit II-               | PROCESSING BIG DATA Introduction to NoSQL – aggregate, key-value and document data models – relationships – graph, schema less databases – materialized views – distribution models – sharding – master-slave and peer- peer replication – consistency & relaxing – version stamps – map reduce – partitioning and combining – composing map-reduce calculations- Analysis of Data format with Hadoop – scaling out – Hadoop streaming & pipes – Design of Hadoop distributed file system (HDFS) and related concepts- Selecting appropriate execution modes: local, pseudo-distributed, fully distributed. | [7 Hrs]      |
| Unit III               | MAP REDUCE APPLICATIONS Employing Hadoop Map Reduce - Creating the components of Hadoop Map Reduce jobs & workflows – unit tests with MRUnit – test data and local tests – anatomy of MapReduce job run – classic Map-reduce – Distinguishing Hadoop daemons - YARN – failures in classic Map-reduce and YARN – job scheduling – shuffle and sort – task execution – MapReduce types – input & output formats                                                                                                                                                                                               | [7 Hrs]      |
| Unit IV                | HADOOPRELATED TOOLS Hbase – data model and implementations – Hbase clients – Hbase examples – praxis.Cassandra – cassandra data model – cassandra examples – cassandra clients – Hadoop integration. Pig – Grunt – pig data model – Pig Latin – developing and testing Pig Latin scripts. Hive – data types and file formats – HiveQL data definition – HiveQL data manipulation – HiveQL queries. Compute the sum of uploaded ,downloaded total transmitted bytes using Pig.Find the minimum value of client MSS.                                                                                          | [8 Hrs]      |
| Unit V                 | R PROGRAMMING R Nults and Bolts -Getting Data In and Out of R- Control Structures and Functions- Loop Functions- Data Manipulation (dplyr, reshape2 packages)- String Operations (stringr package)- Packaging, Debugging and Object Oriented Programming- Graphics and Data Visualization (ggplot2 package) – Clustering- Regression and Classification - swirl Programming.                                                                                                                                                                                                                                | [7 Hrs]      |

| <b>Text Books</b> |                                                                                                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                | Michael Minelli, Michelle Chambers and AmbigaDhiraj, “Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses”, Wiley, 2013. |
| 2.                | . ArvindSathi, “Big Data Analytics: Disruptive Technologies for Changing the Game”, 1st Edition, IBM Corporation, 2012.                                                 |

| <b>Reference Books</b> |                                                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                     | Bill Franks, “Taming the Big Data Tidal Wave: Finding Opportunities in Huge Data Streams with Advanced Analytics”, 1st Edition, Wiley and SAS Business Series,2012 |
| 2.                     | Eric Sammer, “Hadoop Operations”, O'Reilley, 2012.                                                                                                                 |
| 3.                     | . P. J. Sadalage and M. Fowler, “NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence”, Addison-Wesley Professional, 2012                  |

|    |                                                                                 |
|----|---------------------------------------------------------------------------------|
| 4. | E. Capriolo, D. Wampler, and J. Rutherglen, “Programming Hive”, O’Reilly, 2012. |
|----|---------------------------------------------------------------------------------|

| Semester | Course Code | Name of the course               | L | T | P | Credits |
|----------|-------------|----------------------------------|---|---|---|---------|
| III      | MC3E003     | PROFESSIONAL<br>ELECTIVE-II: IOT | 3 | 0 | 0 | 3       |

#### Prerequisites for the course

|    |                                                                                                                                  |
|----|----------------------------------------------------------------------------------------------------------------------------------|
| 1. | The Internet of Things (IoT) is aimed at enabling the interconnection and integration of the physical world and the cyber space. |
| 2. | It represents the trend of future networking, and leads the third wave of the IT industry revolution.                            |

#### Prior Reading Material/useful links

|    |                                                                                                                                                                         |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | <a href="https://www.techtarget.com/iotagenda/definition/Internet-of-Things-IoT">https://www.techtarget.com/iotagenda/definition/Internet-of-Things-IoT</a>             |
| 2. | <a href="https://www.oreilly.com/library/view/programming-the-internet/9781492081401/">https://www.oreilly.com/library/view/programming-the-internet/9781492081401/</a> |
| 3. | <a href="https://www.oreilly.com/library/view/internet-of-things/9780134667577/">https://www.oreilly.com/library/view/internet-of-things/9780134667577/</a>             |

#### Course Outcomes:

| Sr. No | Course Outcome number | CO statement                                                                                |
|--------|-----------------------|---------------------------------------------------------------------------------------------|
| 1      | CO1                   | Describe the evolution of IoT, IoT networking components, and addressing strategies in IoT. |
| 2      | CO2                   | Classify various sensing devices and actuator types.                                        |
| 3      | CO3                   | Demonstrate the processing in IoT.                                                          |
| 4      | CO4                   | Explain Associated IOT Technologies                                                         |
| 5      | CO5                   | Illustrate architecture of IOT Applications                                                 |

#### Syllabus:

| Course Contents |                                                                                                                                                                                                                                                                  | Hours   |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Unit I-         | Basics of Networking: Introduction, Network Types, Layered network models<br>Emergence of IoT: Introduction, Evolution of IoT, Enabling IoT and the Complex Interdependence of Technologies, IoT Networking Components.                                          | [7 Hrs] |
| Unit II-        | IoT Sensing and Actuation: Introduction, Sensors, Sensor Characteristics, Sensorial Deviations, Sensing Types, Sensing Considerations, Actuators, Actuator Types, Actuator Characteristics..                                                                     | [7 Hrs] |
| Unit III        | IoT Processing Topologies and Types: Data Format, Importance of Processing in IoT, Processing Topologies, IoT Device Design and Selection Considerations, Processing Offloading                                                                                  | [7 Hrs] |
| Unit IV         | Associated iot technologies Cloud Computing: Introduction, Virtualization, Cloud Models, Service-Level Agreement in Cloud Computing, Cloud Implementation, Sensor-Cloud: Sensors-as-a-Service. IOT CASE STUDIES Agricultural IoT – Introduction and Case Studies | [8 Hrs] |
| Unit V          | Iot case studies and future trends Vehicular IoT – Introduction Healthcare IoT – Introduction, Case Studies IoT Analytics – Introduction                                                                                                                         | [7 Hrs] |



| <b>Text Books</b> |                                                                                                                                                                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                | Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, “From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence”, 1st Edition, Academic Press, 2014. ( ISBN-13: 978-0124076846) |

| <b>Reference Books</b> |                                                                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                     | Vijay Madisetti and Arshdeep Bahga, “Internet of Things (A Hands-on-Approach)”, 1st Edition, VPT, 2014. ( ISBN-13: 978-8173719547)                                     |
| 2.                     | Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting Everything”, 1st Edition, Apress Publications, 2013. ( ISBN-13: 978-1430257400) |

| Semester | Course Code | Name of the course     | L | T | P | Credits |
|----------|-------------|------------------------|---|---|---|---------|
| III      | MC3L006     | MACHINE LEARNING (LAB) | 2 | 0 | 0 | 2       |

#### Prerequisites for the course

|    |                                                                                                                                                                                          |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | To make the computers smarter, more intelligent. The more direct objective in this aspect is to develop systems (programs) for specific practical learning tasks in application domains. |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Prior Reading Material/useful links

|    |                                                                                                                                                                                         |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | <a href="https://www.simplilearn.com/iitk-professional-certificate-course-ai-machine-learning">https://www.simplilearn.com/iitk-professional-certificate-course-ai-machine-learning</a> |
| 2. | <a href="https://intellipaat.com/machine-learning-certification-training-course/">https://intellipaat.com/machine-learning-certification-training-course/</a>                           |

#### Course Outcomes:

| Sr. No | Course Outcome number | CO statement                                                                                                                                                                   |
|--------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | CO1                   | Apply knowledge of database for real life applications.                                                                                                                        |
| 2      | CO2                   | Apply query processing techniques to automate the real time problems of databases. K3, K4                                                                                      |
| 3      | CO3                   | Identify and solve the redundancy problem in database tables using normalization                                                                                               |
| 4      | CO4                   | Understand the concepts of transactions, their processing so they will familiar with broad range of database management issues including data integrity, security and recovery |
| 5      | CO5                   | Design, develop and implement a small database project using database tools.                                                                                                   |

**Syllabus:**

| <b>List of Experiments:</b> |                                                                                                                                                                                                   |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                           | Study and installation of python                                                                                                                                                                  |
| 1.                          | Study of Python Basic Libraries such as Statistics, Math, Numpy and Scipy                                                                                                                         |
| 2.                          | Write a Python program to implement Simple Linear Regression                                                                                                                                      |
| 3.                          | Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample |
| 4.                          | Implementation of K-Means Clustering                                                                                                                                                              |
| 5.                          | Implementation of KNN using sklearn                                                                                                                                                               |
| 6.                          | Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.                                                                   |
| 7.                          | Performance analysis of Classification Algorithms on a specific dataset (Mini Project)                                                                                                            |
| 8.                          | Implement linear regression using python.                                                                                                                                                         |
| 9.                          | Implement Naïve Bayes theorem to classify the English text                                                                                                                                        |

| Semester | Course Code | Name of the course | L | T | P | Credits |
|----------|-------------|--------------------|---|---|---|---------|
| III      | MC3P007     | MINI PROJECT       | - | 0 | 0 | 2       |

| MINI PROJECT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>➤ General Suggestions and Expectations / Guidelines</li> <li>➤ The project shall be developed in C++/JAVA/PYTHON</li> <li>➤ The students may choose the theory concepts they studied in different subjects as project topic.</li> <li>➤ Interdisciplinary project proposals and innovative projects are encouraged and more appreciable.</li> <li>➤ The project topic can be suggested by the staff member or it can be proposed by the students.</li> <li>➤ The project topic shall be approved by the project in-charge.</li> <li>➤ The Guides are advised to give projects and suggest project titles focusing more on the current field of research and ensure the level of innovation.</li> <li>➤ A project team shall contain</li> <li>➤ The project work should be properly distributed among the team members.</li> <li>➤ Students should submit the project documentation at the beginning of the semester consisting of:</li> <li>➤ Title, Abstract Modules Split-up Deliverables for each review Data Model (If Any) Details of Team Members</li> <li>➤ Reviews for the project work will be conducted at regular intervals by the panel of examiners formed by the Head of the Department.</li> </ul> |

- The student failing to attend the project review will be subject to strict action as decided by the Head of the Department.
- Throughout the semester at any point of time if students are found to be involved in any of the following:
  - Using project codes available on the Internet
  - Using project codes developed by someone else Using project work which is already submitted in other institute or university Such students shall be declared failed or penalized as decided by the Examiners.
- The students must arrange regular meetings with the guide and present progress of project work.
- A Spiral bound Project report to be prepared as per the guidelines and format given by the Department

| Semester | Course Code | Name of the course                 | L | T | P | Credits |
|----------|-------------|------------------------------------|---|---|---|---------|
| III      | MC3L008     | R Programming for Data Science LAB | - | 0 | 0 | 2       |

| Course Outcomes |      |                                                                                           |
|-----------------|------|-------------------------------------------------------------------------------------------|
| Sr No           | Code | CO statement                                                                              |
| 1               | CO1  | Understand and apply basic R syntax and data types for performing operations on datasets. |
| 2               | CO2  | Develop the ability to write basic R scripts and functions to solve problems.             |
| 3               | CO3  | Analyze and visualize data using statistical techniques and basic plotting.               |
| 4               | CO4  | Work with data structures like vectors, matrices, lists, and data frames effectively.     |
| 5               | CO5  | Import, manipulate, and export datasets, preparing them for further data analysis.        |

| List Of Experiments |                                                                                                                                                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sr No               | Description                                                                                                                                                                                                                               |
| 1                   | <b>Introduction to R:</b> <ul style="list-style-type: none"> <li>Install R and RStudio.</li> <li>Write a simple R script that prints "Hello, World!" and outputs basic mathematical operations.</li> </ul>                                |
| 2                   | <b>R Data Types:</b> <ul style="list-style-type: none"> <li>Create variables of different data types (numeric, character, logical).</li> <li>Check the class and type of each variable and convert them to another data type.</li> </ul>  |
| 3                   | <b>Working with Vectors:</b> <ul style="list-style-type: none"> <li>Create numeric and character vectors.</li> <li>Perform operations like indexing, slicing, and arithmetic on vectors.</li> </ul>                                       |
| 4                   | <b>Lists and Matrices:</b> <ul style="list-style-type: none"> <li>Create lists and access elements within them.</li> <li>Create matrices and perform matrix operations (addition, subtraction, multiplication).</li> </ul>                |
| 5                   | <b>Data Frames:</b> <ul style="list-style-type: none"> <li>Create a data frame with student information (name, age, grades).</li> <li>Access specific rows and columns, and summarize the data.</li> </ul>                                |
| 6                   | <b>Control Structures:</b> <ul style="list-style-type: none"> <li>Write an R program using if-else and for loops to calculate the factorial of a number.</li> </ul>                                                                       |
| 7                   | <b>R Functions:</b> <ul style="list-style-type: none"> <li>Define custom functions in R.</li> <li>Write a function to calculate the square of a number and return the result.</li> </ul>                                                  |
| 8                   | <b>Basic Statistics with R:</b> <ul style="list-style-type: none"> <li>Create a dataset and calculate mean, median, mode, and standard deviation.</li> <li>Visualize the data using basic plots like histograms and bar plots.</li> </ul> |
| 9                   | <b>Importing and Exporting Data:</b> <ul style="list-style-type: none"> <li>Import data from CSV and Excel files into R.</li> <li>Export a data frame into a CSV file.</li> </ul>                                                         |
| 10                  | <b>Basic Data Visualization:</b> <ul style="list-style-type: none"> <li>Use the ggplot2 library to create a scatter plot and line graph.</li> <li>Customize the plot with labels, titles, and colors.</li> </ul>                          |

| Content Beyond Syllabus |                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Sr No                   | Description                                                                                                 |
| 1                       | <b>Basic Introduction to Tidyverse:</b> Learn how to use the Tidyverse package to clean and transform data. |
| 2                       | <b>Handling Missing Data in R:</b> Techniques to detect and handle missing values in datasets.              |